Anuj Patel

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WORK EXPERIENCE

Dune Security **Dec 2025 – Present**
Senior AI/ML Engineer New York, NY

- Architected **red-team-agent**, a **stateful agent loop** (observe → plan → act) with dual LLMs, persistent session state, and a **6-state FSM**; integrated **4 channel gateways** (WhatsApp, SMS, Viber, Telegram), targeting **100K** users.
- Built **persona-service** (Go microservice): end-to-end **Claude Sonnet 4.6** powered AEP generation, Groq test harness, and versioned S3 publish; **99%** faster persona builds and **10×** campaign throughput at current headcount.
- Created **vishingService**, a live voice red-team agent (**KAG pipeline**: Groq + Cartesia STT/TTS, Telnyx) with **525ms p50 turn latency** (user stop → first audio) and goal-weighted call scoring.
- Prototyped **Knowledge Faceoff** (**$500K** revenue target, **11%** MAU growth): a gamified security training platform with LLM-generated quizzes from 400+ training transcripts; targets **79%** lower phishing failure rate (**6.56%** → **1.38%**).
- Modeled **customer risk-score decrease**: validating median reduction from **48→38** over 1 year; used in **enterprise renewals** and investor conversations.

New York University **Jan 2024 – Dec 2024**
AI/ML Engineer New York, NY

- Productionized a **RAG-powered chatbot** using **LangChain**, **OpenAI APIs**, and **Pinecone**, enabling natural language search over 100+ academic and policy documents with **sub-300ms** median latency .
- Engineered **retrieval and recommendation pipelines** using **text-embedding-ada-002**, **FastAPI**, and **PostgreSQL** in a **Docker CI/CD** environment, powering course recommendations for **5K+ students**.

Johnson & Johnson **Jun 2024 – Aug 2024**
Data Scientist New Brunswick, NJ

- Led development of a clinical assistant using **LLaMA 2–7B** with **QLoRA and PEFT** over 1M+ anonymized patient records; reducing query time by **46%** across **450+ daily clinical queries** during pilot phase.
- Built a **real-time, multimodal ML pipeline** (text, imaging, vitals) using **PyTorch, HuggingFace** and **AWS**; achieving **28% lift** in outcome prediction accuracy and scaling to 20M+ patient records.

EDUCATION

New York University **Sep 2023 – May 2025**
Master of Science in Electrical Engineering **(GPA: 3.9/4.0)** New York, NY

- **Co-authored** an engineering textbook on **"Fundamentals of Communication Theory"** with **Dr. Unnikrishna Pillai**.
- Coursework: ML, Deep Learning, CV, High Performance Machine Learning, Probability, Big Data
- Research: Developed a mmWave channel sounder at 57.51 GHz under **Dr. Sundeep Rangan** for wireless channel measurements.

PROJECTS

Multi-Agent RAG Chatbot for E-commerce Customer Support **Aug 2025 – Sep 2025**

- Designed an **agentic RAG chatbot** with **7 tools** and **Voyage-3.5 embeddings**, handling **20K+ parts** via dual-database orchestration (SQLite FTS5 + ChromaDB) in **2 LLM calls**.
- Implemented a **multi-agent workflow** with RAGOrchestrator for **intent classification** and **semantic retrieval**, enabling real-time troubleshooting and parts compatibility.

Efficient Federated Learning using Gradient Pruning and Adaptive Methods **Sep 2024 – Dec 2024**

- Pioneered a FL framework that **reduced training time by 22%** and boosted generalization using adaptive optimization.
- Achieved **143% bandwidth efficiency gain** via gradient compression and mixed-precision training with **PyTorch DDP, DeepSpeed, and HF Accelerate**.

Transformer-Based Multi-Modal Emotion Recognition System **Sep 2024 – Dec 2024**

- Enhanced a transformer for emotion recognition, achieving **33.9% top-1** and **98.1% top-5 precision** on the RAVDESS dataset.
- Applied advanced **fusion techniques** over **4K+ video-audio samples**, improving robustness in noisy and incomplete datasets.

SKILLS

Languages	Python, Go, TypeScript, C/C++, CUDA, SQL, Bash
Frameworks	PyTorch, HuggingFace, vLLM, FastAPI, LangChain, LangGraph, Numpy, Pandas, Wandb
Cloud	AWS (SageMaker, EC2, ECS, EB, S3, Redshift, Lambda, ECR, CloudWatch)
DevOps	Docker, Kubernetes, Git, CI/CD, Kubeflow, Airflow, Spark, Kafka, Slurm
Databases	PostgreSQL, MongoDB, DynamoDB, SQLite, ChromaDB, Weaviate, Pinecone